

Apache Cassandra Data Model: Part 1

Logical entities: Tables and keyspaces

Table

- Is the logical entity that organizes data storage at cluster and node level according to a declared schema

Keyspace

- Is the logical entity that contains one or more tables
- Replication and data centers' distribution are defined at the keyspace level
- Recommended 1 keyspace per application

- Cassandra stores data in tables whose schema defines the storage of the data at cluster and node level. Tables are grouped in key spaces. A keyspace is a logical entity that contains one or more tables. A keyspace also defines a number of options that applies to all the tables it contains, most prominent of which is the replication strategy used by the key space. It is generally encouraged to use one keyspace per application.

Logical entities: Tables

- Data is organized in tables containing rows of columns
- Tables can be created, dropped, and altered at runtime without blocking updates and queries
- To create a table, you must define a primary key and other data columns (regular columns)

```
CREATE TABLE intro_cassandra.groups(  
  groupid int,  
  group_name text STATIC,  
  username text,  
  age int,  
  PRIMARY KEY ((groupid), username));
```

Partition Key

Clustering Key

- So tables are the logical entities that organize data storage at cluster and node level. They contain rows of columns. You can create, drop, and alter your tables without impacting the running updates on your data or the running queries. In order to create a table, we need to declare using Cassandra query language a schema.
- A table schema comprises at least a definition of the table's primary key and the regular columns of the table. Table groups store information regarding several groups, such as groupid, group_name, and for each group, the username and age of their members. You can see that the primary key is composed of two columns, groupid and username. In Cassandra denomination, the groupid column is called the partition key and the username column is called the clustering key. Let's talk a bit more about the primary key.

Primary key in Cassandra tables

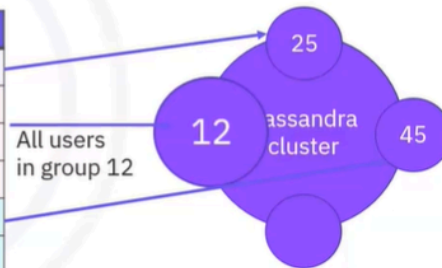
- Is a subset of the declared columns
 - Is mandatory: You cannot change the key once defined
 - Performs two main roles:
 - Optimizes the read performance for your queries on the table: Query-driven table design
 - Provide uniqueness to the entries
 - Has two components:
 - Partition key is mandatory
 - Clustering keys are optional
-
- First of all, the primary key is basically a subset of the table's declared columns. When creating a table, besides declaring the table's columns, it is mandatory to specify the primary key as well. The primary key, once defined, cannot be changed.
 - In Cassandra, the primary key has two roles. The first role is to optimize the read performance of your queries. Do not forget that the NoSQL systems are query driven data modeling systems, meaning that table definitions can start only after the queries you would like to answer are defined. You should build your primary key based on your queries, and the second role is to provide uniqueness to the queries. A primary key has two components. The mandatory component is called the partition key, and optionally you can have one or more clustering keys.

Partition keys

- When inserting data into a table, data is grouped into partitions and distributed on the cluster nodes based on the partition key
- Partition key => Hash (token) => Node
- The Partition key determines data (partition) locality in the cluster

GroupID	Group_name	Username	Age
25	Vegan cooking	Johns@gmail.com	60
12	Baking	Alaind@gmail.com	32
12	Baking	Elaine@yahoo.com	60
12	Baking	Peterd@gmail.com	32
45	Grilling	Moirad@yahoo.com	35
45	Grilling	Daveg@gmail.com	43

PRIMARY KEY ((groupid), username))



- When data is inserted into the cluster in a table, the data is grouped per partition key into partitions, and the first step is to apply a hash function to the partition key. The partition key hash is used to determine what node and subsequent replicas will get the data. In simpler terms, a partition key determines the data locality in the cluster.
- You can see in the diagram and table that data is grouped according to the partition key, groupid, and that each partition is distributed to one of the cluster nodes. The partition is the atom of storage in Cassandra, meaning that one partition's data will always be found on a node and its replicas in the case of a replication factor greater than one.
- So if we want to answer the query all users in group 12, then the query can address only the fourth node and will get the answer. In big clusters, those consisting of hundreds or thousands of nodes, limiting the number of nodes required to be contacted in order to answer the query is crucial for query performance.

Table types

Two types of tables: Static and dynamic

- Static table
 - PRIMARY KEY (username)
- Dynamic table
 - PRIMARY KEY ((groupid), username))

- Let's define two more concepts, static and dynamic tables. When the table has a primary key that contains only the partition key, single or multiple columns, and no clustering key, the table is called a static table. When the table has the primary key composed of both partition key or keys, and clustering key or keys, the table is called a dynamic table.

Static tables

Users: Static table
Partition key = username
No Clustering key
1 partition = 1 entry

Username	Occupation	Age
Johns@gmail.com	Engineer	60
Alaind@gmail.com	Programmer	32
Elaine@yahoo.com	Engineer	60
Peterd@gmail.com	Marketer	27
Moirad@yahoo.com	None	35
Daveg@gmail.com	Pianist	43
JayZ@yahoo.com	Entertainer	46

```
CREATE TABLE
intro_cassandra.users(
  username text,
  occupation text,
  age int,
  PRIMARY KEY (username));
```

- This user's table is a static table. As you can see, the primary key consists only of the partition. This means that the number of distinct users we will have is the number of partitions we will have in our table. In static tables, partitions have only one entry and thus are called static partitions.

Recap

In this video, you learned that:

- Cassandra stores data in tables, and tables are grouped in keyspaces
- It is recommended to use one keyspace per application
- You can create, drop, and alter your tables without impacting the running updates or queries on your data
- The two roles of a primary key are to optimize the read performance for your queries on the table and to provide uniqueness to the entries
- A primary key has two components: a mandatory partition key and one or more optional clustering keys
- Cassandra has two types of tables: Static and dynamic
- Static tables are those tables that have a primary key that contains only the partition key but no clustering key
